In-Context Grammar Learning in Mistral-7B: Full Results

Nov 9, 2026 · @Dor

Summary

A frozen Mistral-7B learns two artificial grammars from its prompt alone. This learning becomes readable abruptly at **layer 19**, and it is produced mainly by a small group of **induction heads in block 18**.

Test	Question	Main result
1. Reproduction	Does the paper's result hold?	Yes: AGL reaches ~85% of the bound, SRTT near 100%; all controls behave as reported.
2. Logit lens per layer	At which layer does the learning become visible?	Abruptly at layer 19 (AGL 2% → 40% of the bound in one layer), small step at 13.
3. Tuned lens	Is the jump an artifact of how we read layers?	No: a better-trained reader shows the same jump at 19.
4. Induction-head ablation	Which component produces the jump?	Switching off 16 of 1,024 heads halves AGL learning (0.75 → 0.37), cuts SRTT 28%, erases the jump. Random heads: no effect.
5. Noise per layer	Where is the learning most fragile?	Noise just before block 18 (layers 15–18) hurts most; after 19, almost nothing. Weak effect.
6. Head specialization	Does each head handle different transitions?	SRTT: heads are interchangeable copies. AGL: heads combine, split into "end" vs. "continue".

Setup (shared by all tests)

Model: Mistral-7B-v0.3, frozen (no weight updates), loaded in 8-bit. It learns only from what it reads in the prompt.

Word streams. The model reads one long stream built from 4 random English words plus `END`. At every position we record the probability it gave to the word that actually came next. Three parts:

1. **Random** (~420 words): no rules, so the best guess is 1 in 5 = 0.2.
2. **Grammar** (~2,450 words for AGL, 2,880 for SRTT): the same words now follow rules.
3. **Random again** (~410 words): a control. If the model learned the rules, its advantage disappears here.

Two grammars (classic human statistical-learning tasks):

- **SRTT** (Serial Reaction Time Task): one fixed 8-word sequence plus END, repeated, like a song on repeat. Fully predictable, so the ideal score is 1.
- **AGL** (Artificial Grammar Learning): a 5-state rule system (q0–q4) where each next word is chosen at random among 2–3 allowed options. The rules are learnable, the choice is not, so the ideal score is about 0.46.

Every run uses a different random vocabulary, so nothing depends on particular words (100 runs in Tests 1–3, 50 in Test 4, 10–20 in Tests 5–6).

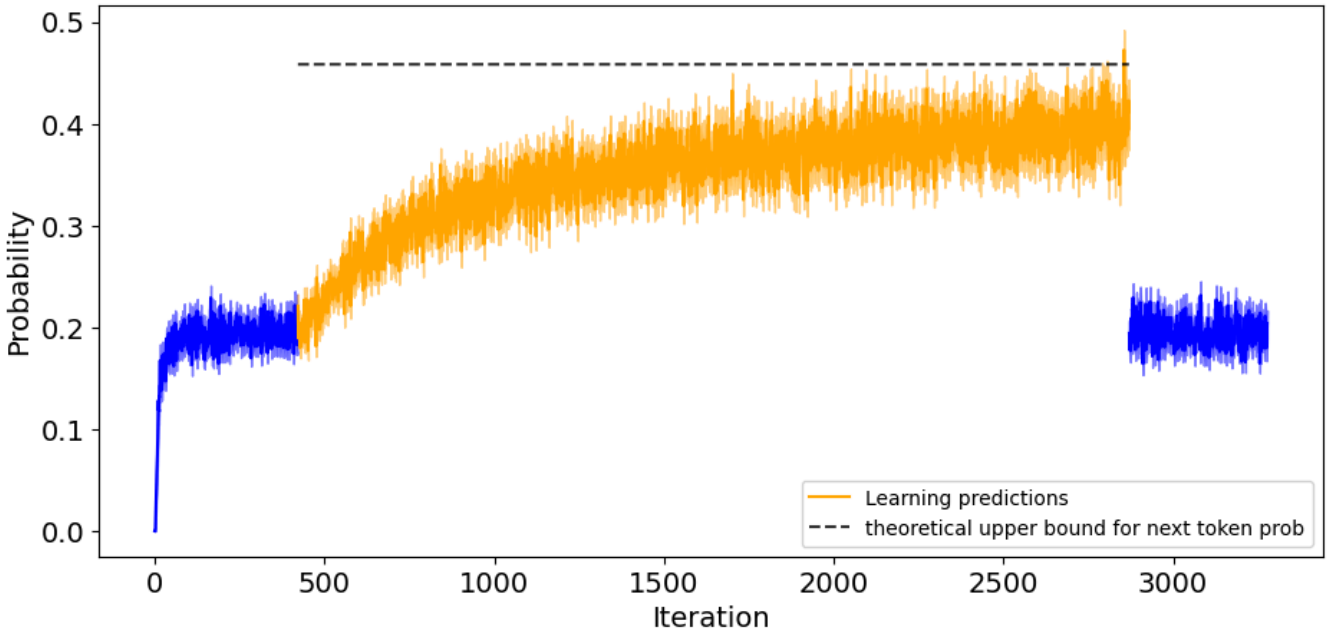
Learning score (used from Test 2 on): (phase mean − baseline) ÷ (bound − baseline).

- **Phase mean:** average probability of the correct next word over the last 1,000 grammar words, when learning is mostly done.
- **Baseline:** the same average over the first random part (skipping the first 50 warm-up words).
- **Bound:** the best a perfect learner could score (0.46 for AGL, 1.0 for SRTT).

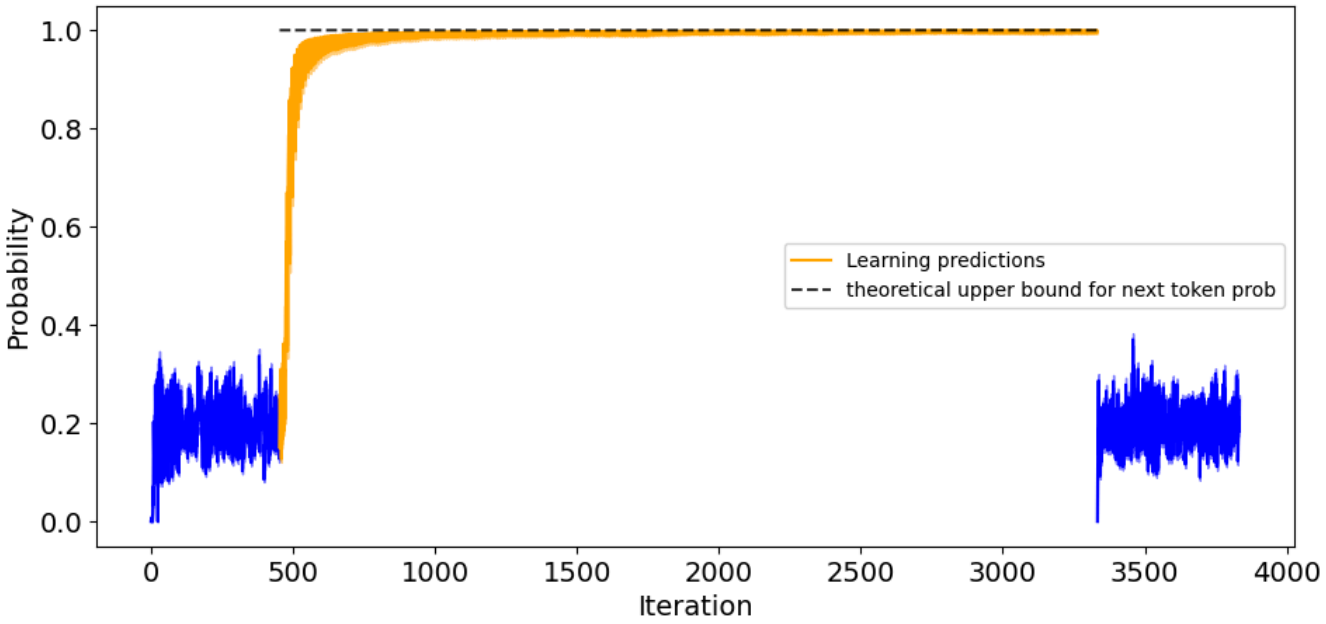
0 = no better than in the random part; 1 = a perfect learner.

Test 1: Reproducing the paper

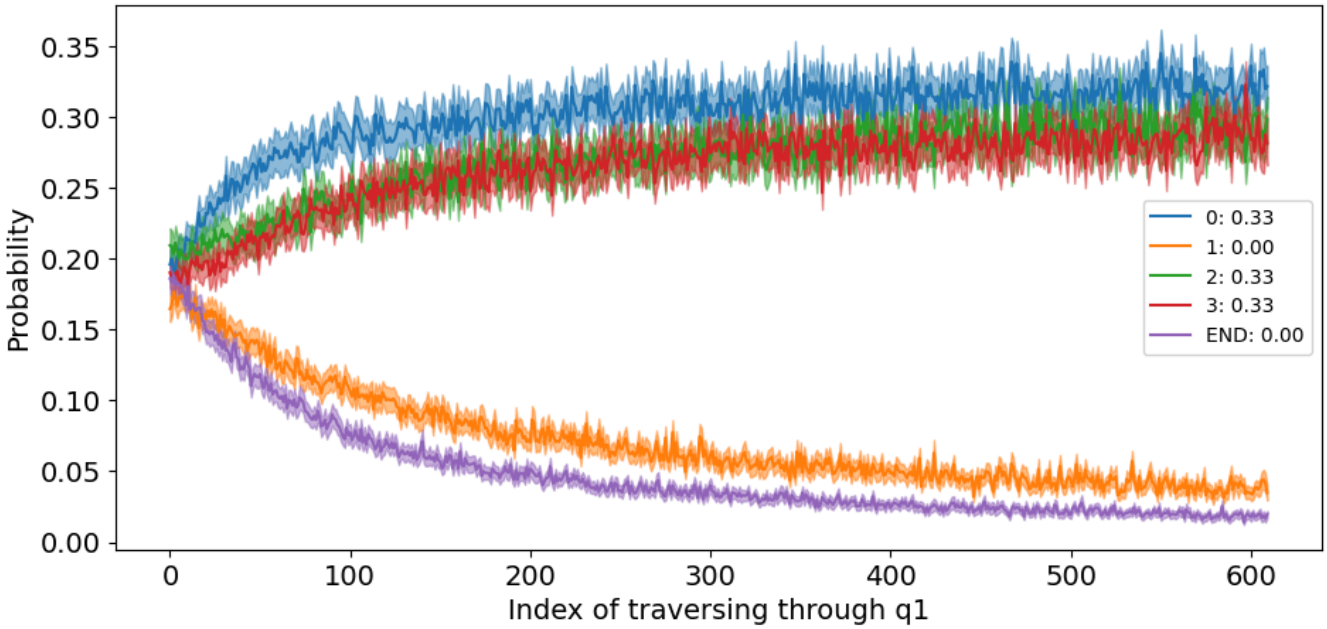
Question: does a frozen Mistral-7B really learn the grammars from context alone?
Answer: yes; the main results match the paper. AGL reaches about 85% of the bound (0.39 of 0.46; paper: 79%), SRTT is near perfect (paper: 92%). These plots measure only the model′s final output, as in the paper.



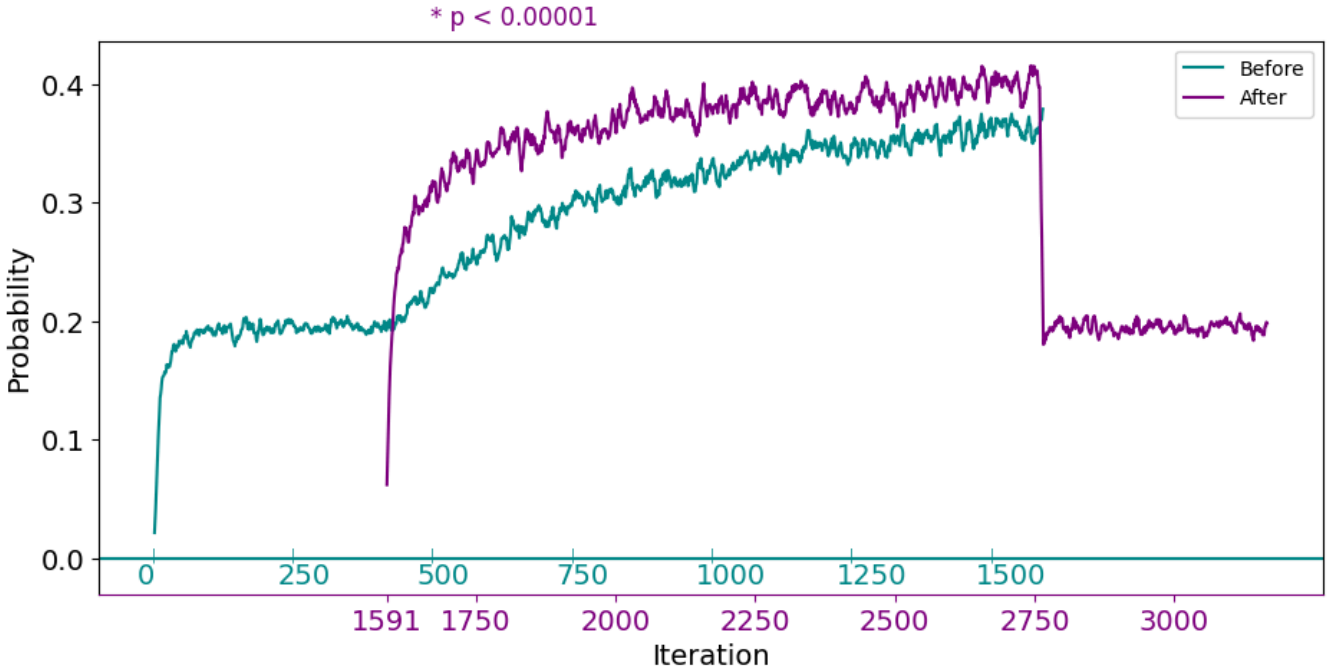
AGL. At chance (0.2) in the random part, climbs to ~0.39 during the grammar (bound 0.46, dashed), and drops straight back to 0.2 when the rules disappear.



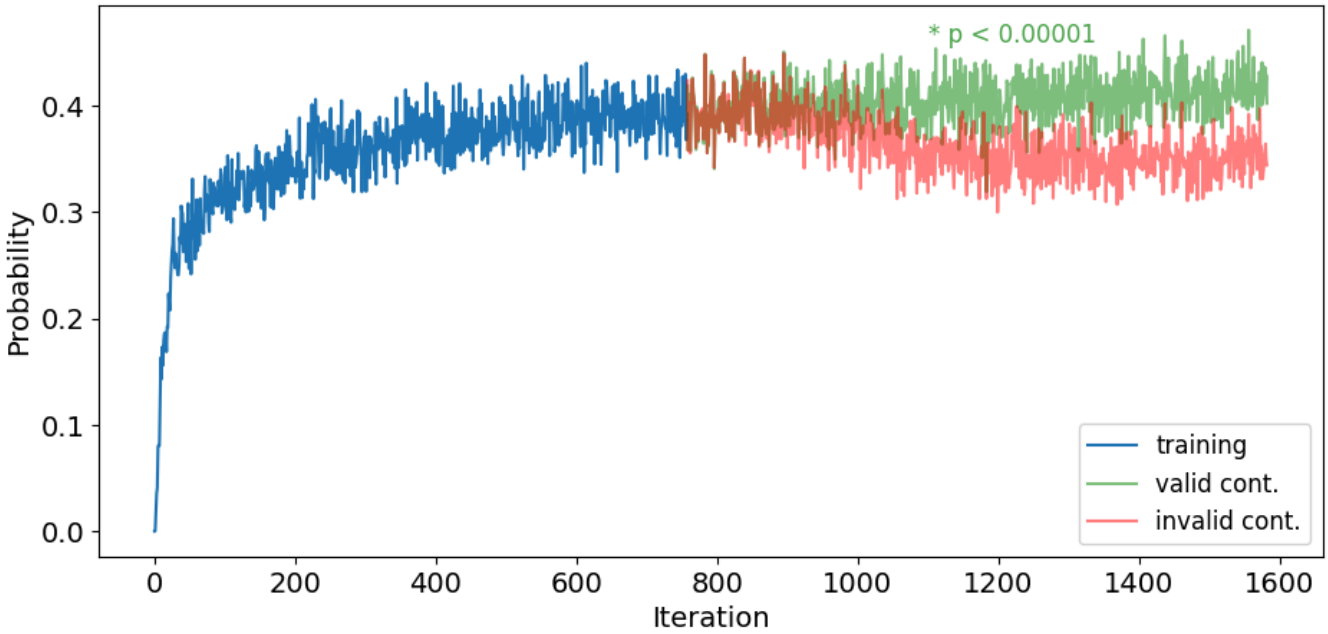
SRTT. Reaches ~0.95 within about 10 repetitions and ~0.99 by the end; back to 0.2 afterwards.



It learns the full rule structure. At state q1, three words are allowed (0.33 each) and two are forbidden. The allowed ones rise toward 0.3; the forbidden ones fall toward 0.02–0.04.



It learns the rules, not the words. Halfway through, the same rules continue with 4 new words. The second time (purple), the model reaches ~0.3 within ~100 words; the first time (teal) it needed ~800.

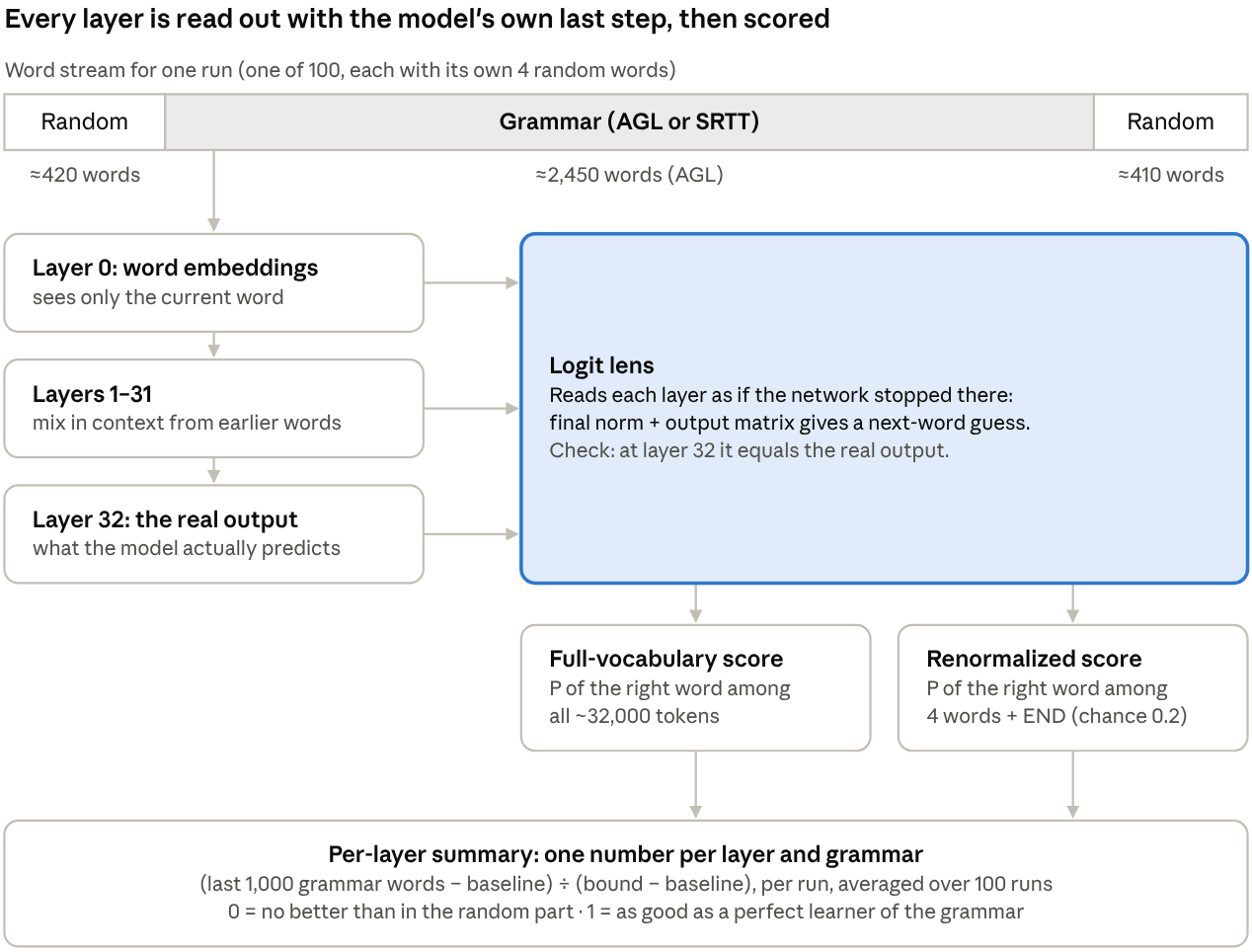


It generalizes and notices violations. New, unseen valid sequences score ~0.41; the same sequences with two neighboring words swapped score ~0.35.

Test 2: Per-layer decoding with the logit lens

Question: the paper looks only at the final answer. Inside, each word passes through 32 layers. *If the model had to answer at layer X, how good would its guess be?* **Answer:** the learning becomes visible abruptly at **layer 19**, for both grammars.

Method (logit lens). The model's last step turns its internal state into a guess over words (a normalization, then the output matrix). We apply that same step to the internal state after **each** layer, as if the network stopped there. Nothing is trained or changed. Lens layer 0 = the word embedding; lens layer L = the output of block L−1.

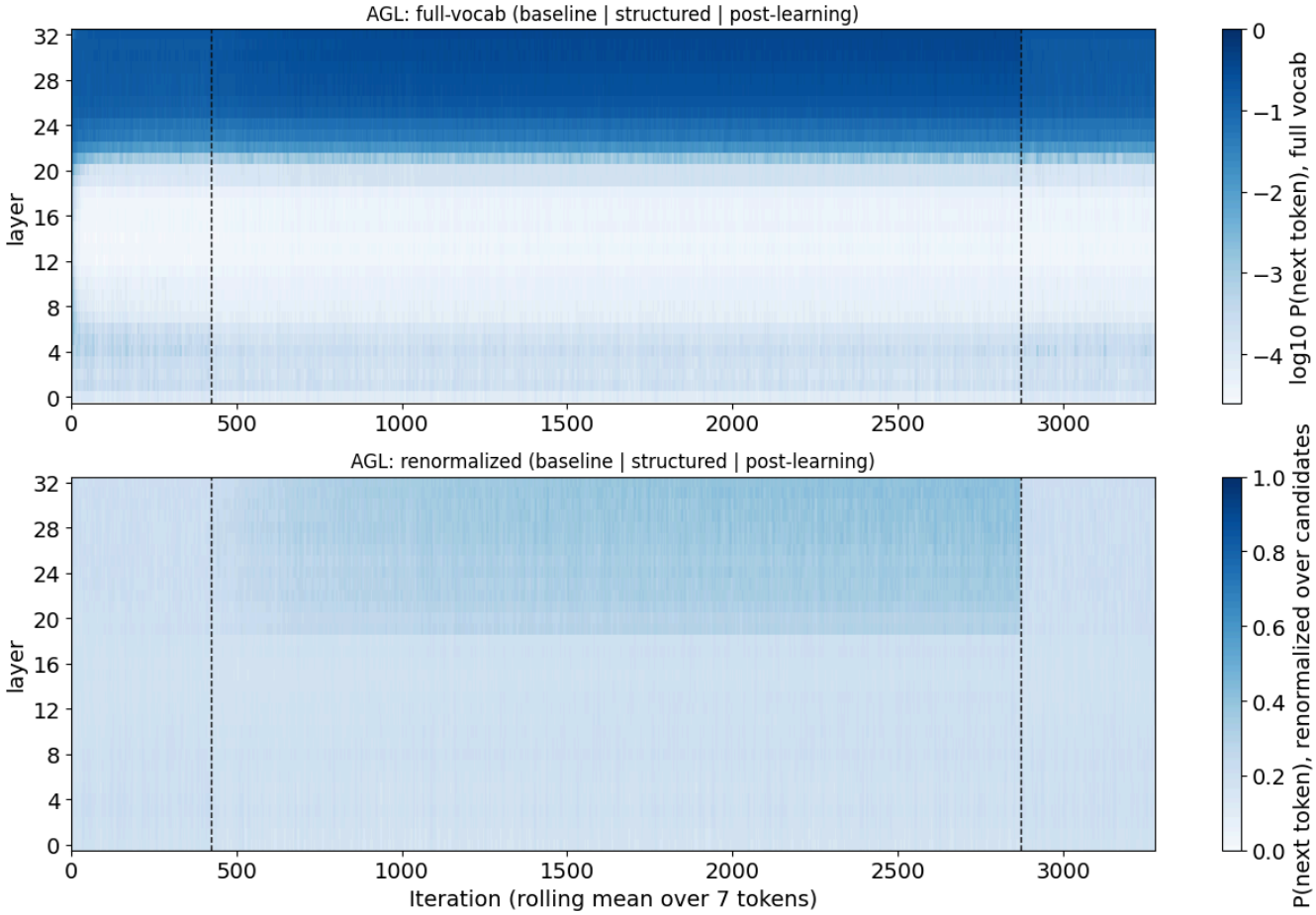


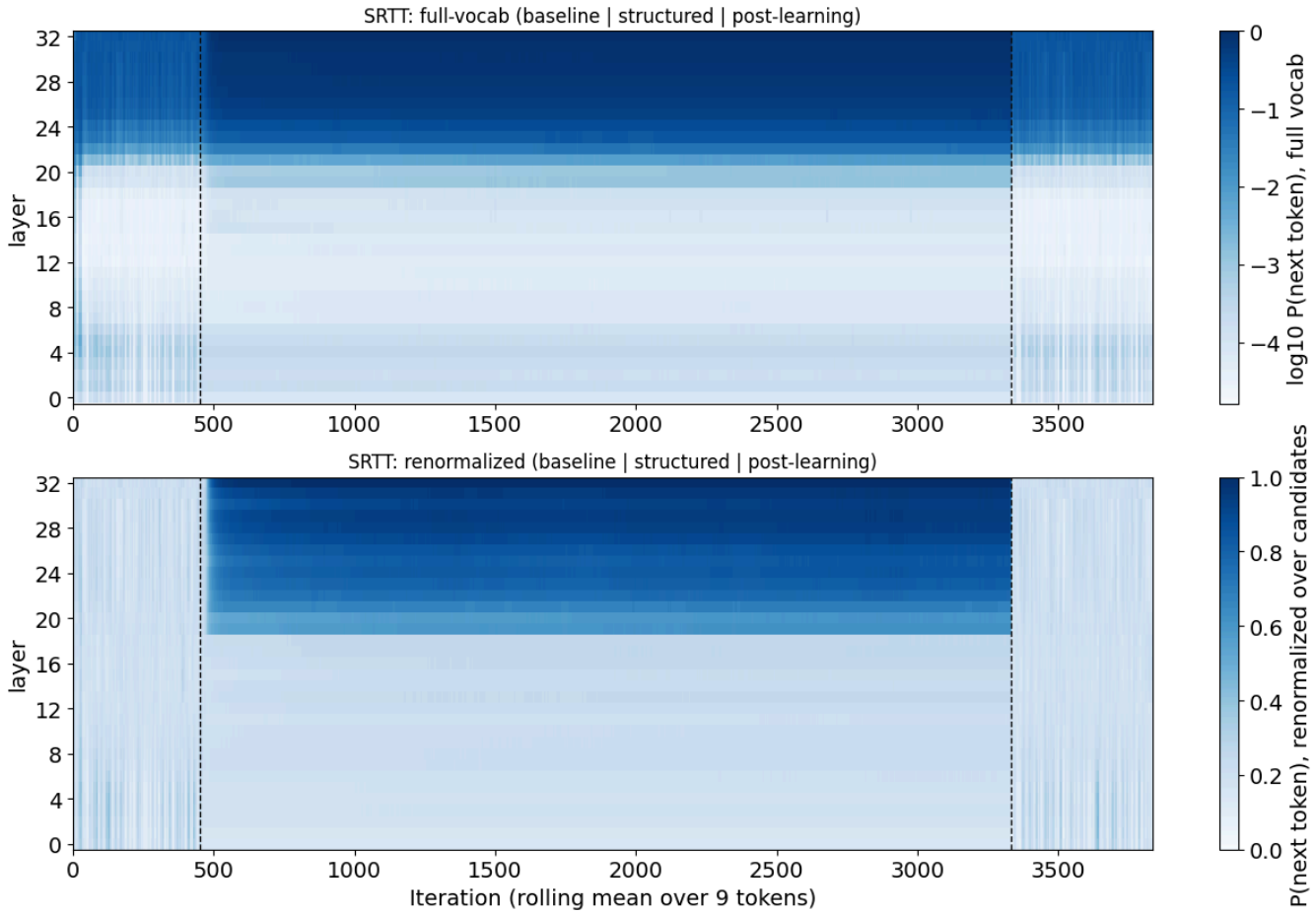
Two scores per layer:

- **Full-vocabulary:** probability of the correct word out of all ~32,000 tokens. What the model would actually say.
- **Renormalized:** among only the 5 possible answers (4 words + END), the share that goes to the correct one. Chance = 0.2. It asks whether a layer already *prefers* the right answer before it can output it.

Example (AGL, renormalized)	Baseline	Phase mean	Bound	Calculation	Score
Layer 19	0.204	0.305	0.458	0.101 ÷ 0.254	0.40
Layer 32	0.206	0.405	0.458	0.199 ÷ 0.252	0.79

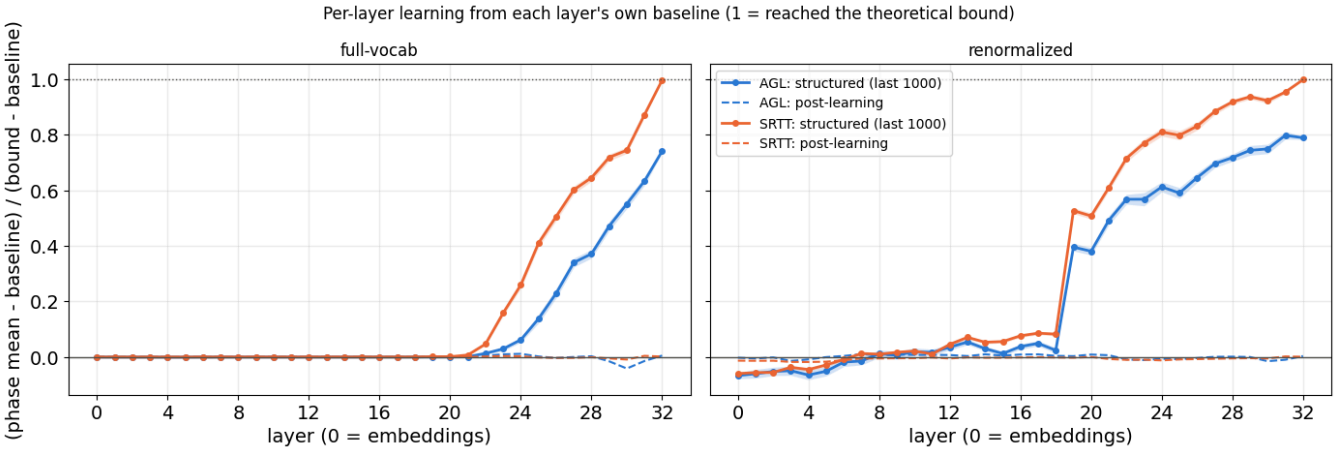
Checks that passed: layer 32 through the lens reproduces the real output exactly; layer 0 (sees only the current word) shows no drift; in the random part every layer scores ~0.19–0.22 (chance).


AGL: full-vocab (baseline | structured | post-learning)
layer
log10 P(next token), full vocab
AGL: renormalized (baseline | structured | post-learning)
layer
P(next token), renormalized over candidates
Iteration (rolling mean over 7 tokens)



Rows = layers (0 bottom, 32 top), columns = positions; dashed lines separate random | grammar | random; darker = higher probability of the correct next word.

- **Layers 0–18** look the same in all three parts: no learning.
- **From layer 19 up,** the grammar part darkens as more examples are read, then fades back to chance when the rules stop.
- **Full-vocabulary** shows learning only from layer ~22: the model "knows" before it can "say".



Layer	AGL, renormalized	SRTT, renormalized	AGL, full-vocab	SRTT, full-vocab
0–11	−0.06 to 0.02	−0.06 to 0.02	0	0
12–18	0.02–0.05	0.05–0.09	0	0
19	**0.40**	**0.53**	0	0
24	0.61	0.81	0.06	0.26
32	0.79	1.00	0.74	1.00

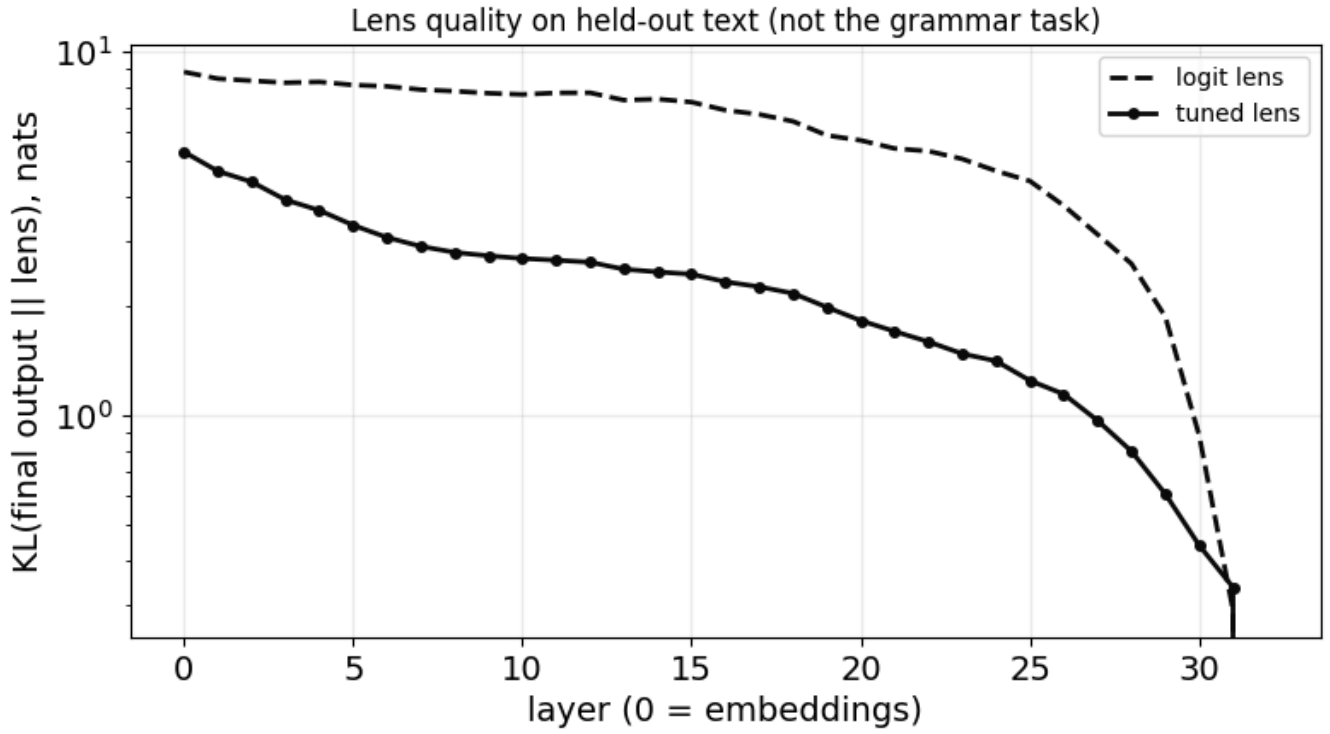
- **One-layer jump at 19** for both grammars (AGL 2% → 40%, SRTT 8% → 53%), then steady improvement up to 32.
- **Same onset and shape for both grammars:** the simple sequence is not resolved earlier than the probabilistic rules.
- **Controls (dashed, random part after the grammar) stay at ~0** at every layer.

Caveat. The lens uses a final step trained for the last layer only, so it may miss information held in another form. The claim is "readable at layer X", not "computed at layer X"; Test 3 addresses this.

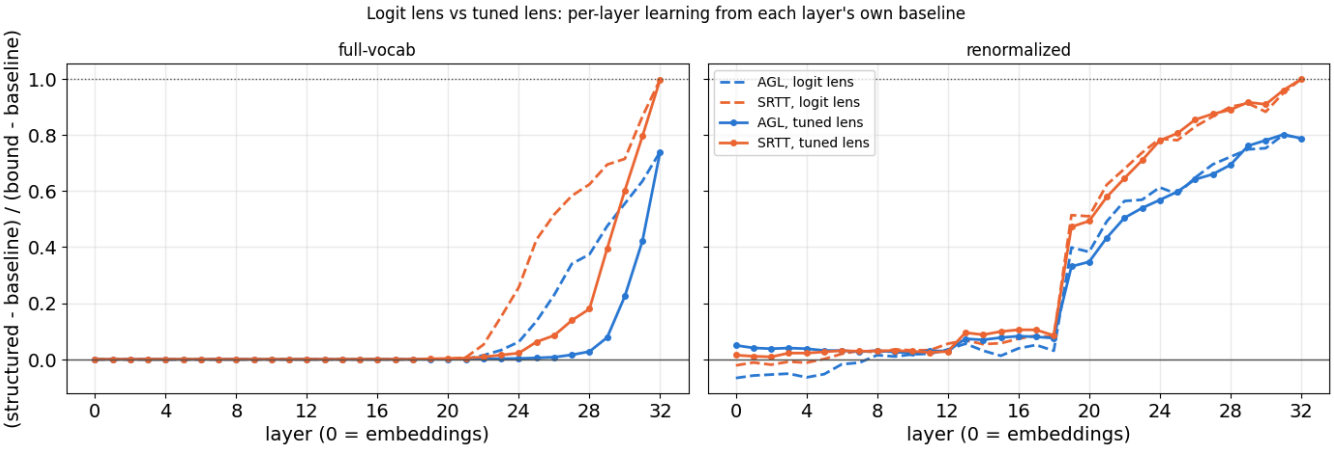
Test 3: Checking with a tuned lens

Question: the logit lens reads every layer with the last layer's translator. Is the jump at 19 real, or an artifact of how we read the layers? **Answer:** real; a better reader shows the same jump.

Method. A tuned lens (Belrose et al., 2023) learns, for each layer, a small linear correction (one matrix plus an offset) that converts the layer's state into the last layer's format. It is trained to match the model's final output on Wikipedia (~400k tokens), never on the grammar, so it can only translate, not learn the grammar. Same runs, scores and summary number as Test 2.



Lens quality. On held-out Wikipedia, the tuned lens is much closer to the real output than the logit lens up to layer 30 (layer 15: 2.4 vs. 7.3 nats; lower is better).



- **Renormalized (right): the two lenses nearly overlap.** Same jump at 19, same rise to 32. The jump is a property of the model, not of the logit lens.
- **A small step at layer 13** appears with both lenses and both grammars (~0.07–0.10 of the bound, flat until 18): a weak early trace.
- **Full-vocabulary (left): the tuned lens rises later,** as expected: trained on English, it favours ordinary words over copying the 4 random context words. For this task the renormalized score is the relevant one.

Caveats. Trained on ~400k tokens (the paper used ~65M), so not fully converged; at layer 31 it is slightly worse than the logit lens (0.34 vs. 0.29 nats). Both lenses are linear and aimed at next-word prediction.

Test 4: Switching off induction heads

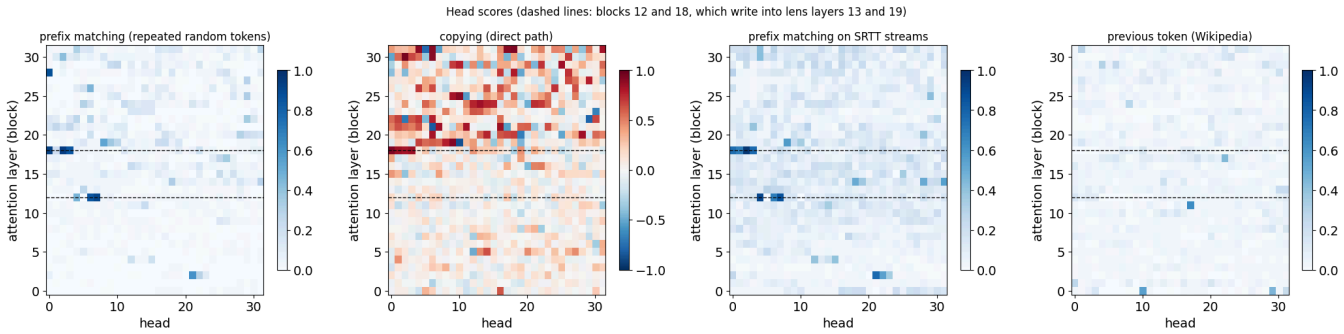
Question: which component produces the jump at layer 19? **Answer:** induction heads in block 18. Switching off 16 of 1,024 heads halves AGL learning and erases the jump; 16 random heads do nothing.

What an induction head is. An attention head that implements [A][B] … [A] → [B] (Olsson et al., 2022): when the current word A appeared earlier, it looks at the word that followed it (B) and copies it into the prediction. It needs only the context, so it is a natural candidate for in-context learning.

Step 1: find the heads. Every head (32 layers × 32) gets two scores, and heads are ranked by their product:

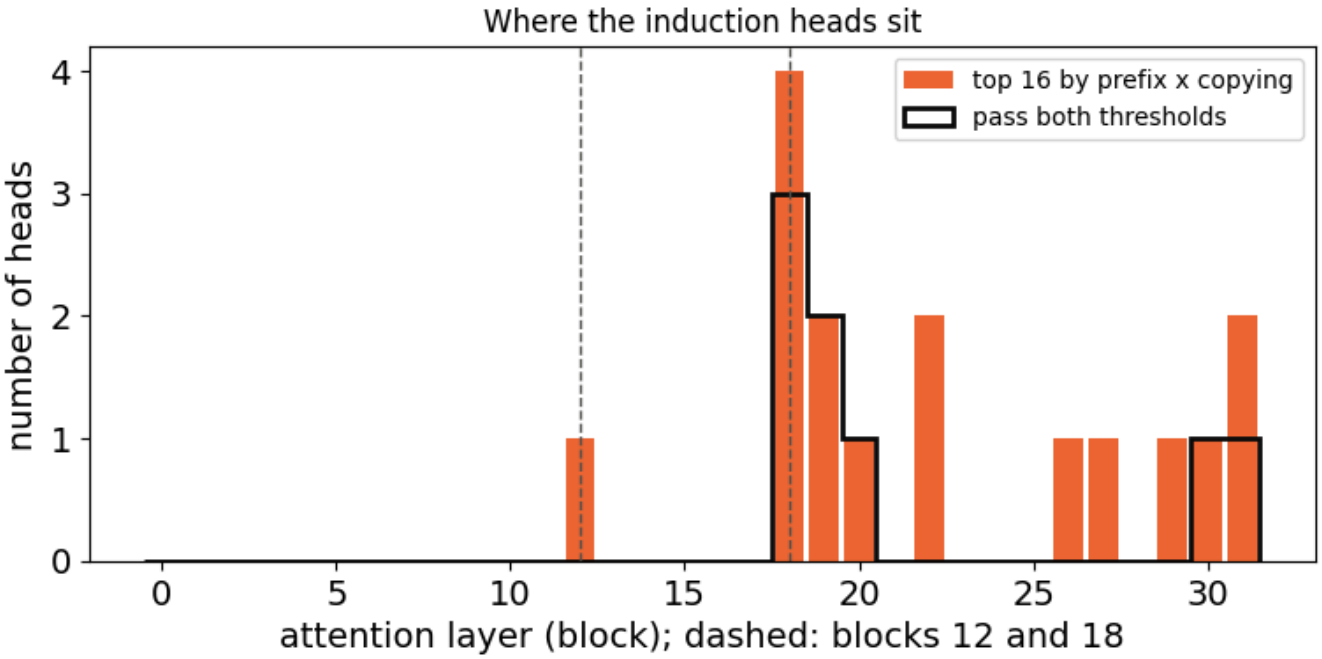
- **Prefix matching** (does it *look* like one?): on 25 random tokens repeated 4 times, the share of attention on the token right after the earlier copy of the current token (0–1).
- **Copying** (does it *act* like one?): the head's output alone, translated into words; where the attended word ranks among all ~32,000 words (+1 = pushed up most, 0 = chance).

Step 2: switch them off (mean ablation). Each head writes 128 numbers into its layer's output; we replace only those with their average over Wikipedia, so the head carries no information about the current input. Conditions: no ablation; top 4, 8, 16; and **16 random heads with the same number per layer** (2 draws), the key control. General-ability checks: Wikipedia loss and Olsson's ICL score (loss at token 500 − at token 50). 50 runs per grammar.

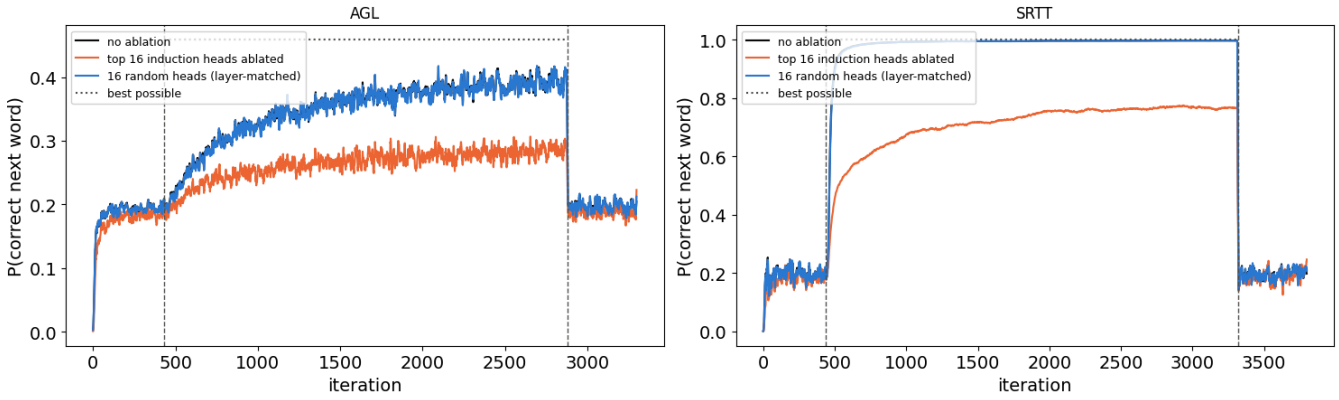


- **Two clusters of strong prefix matching (0.85–0.93):** block 18 (heads 0, 2, 3) and block 12 (heads 6, 7). Block 18 writes lens layer 19 and block 12 writes lens layer 13: exactly the jump and the small step from Tests 2–3.

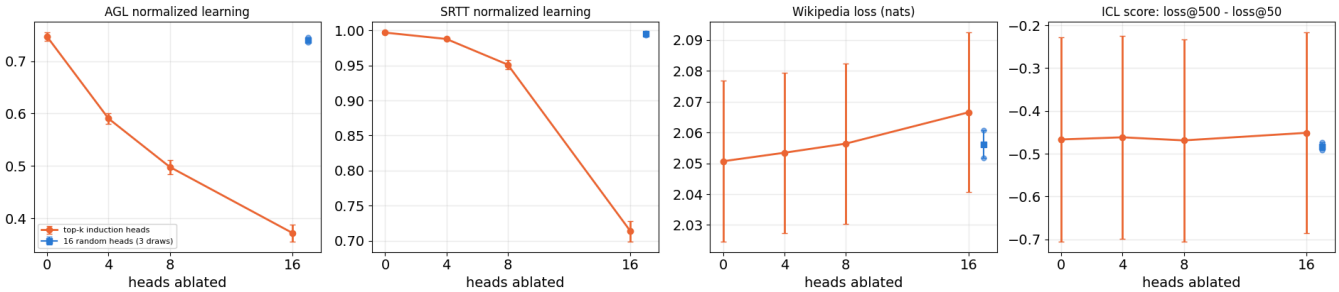
- **The block-18 heads also copy strongly (0.83–0.93):** full induction heads. The block-12 heads copy weakly (~0.2): they likely feed later layers.
- The same heads light up on our own SRTT streams (r = 0.77).



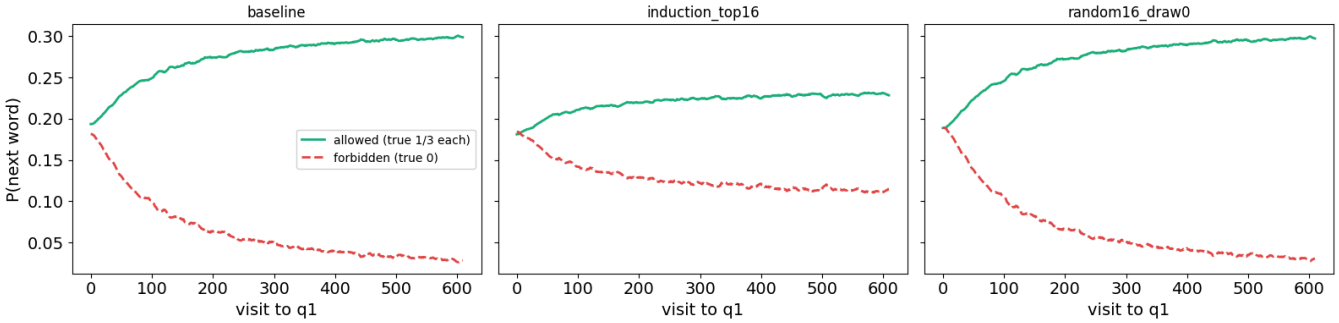
Condition	AGL learning	SRTT learning	Wikipedia loss	ICL score
No ablation	0.747	0.997	2.051	−0.467
Top 4 induction heads	0.590	0.988	2.053	−0.462
Top 8	0.498	0.951	2.056	−0.469
Top 16	**0.372**	**0.713**	2.067	−0.451
16 random heads, same blocks	0.736–0.745	0.993–0.997	2.052–2.061	−0.474 to −0.491



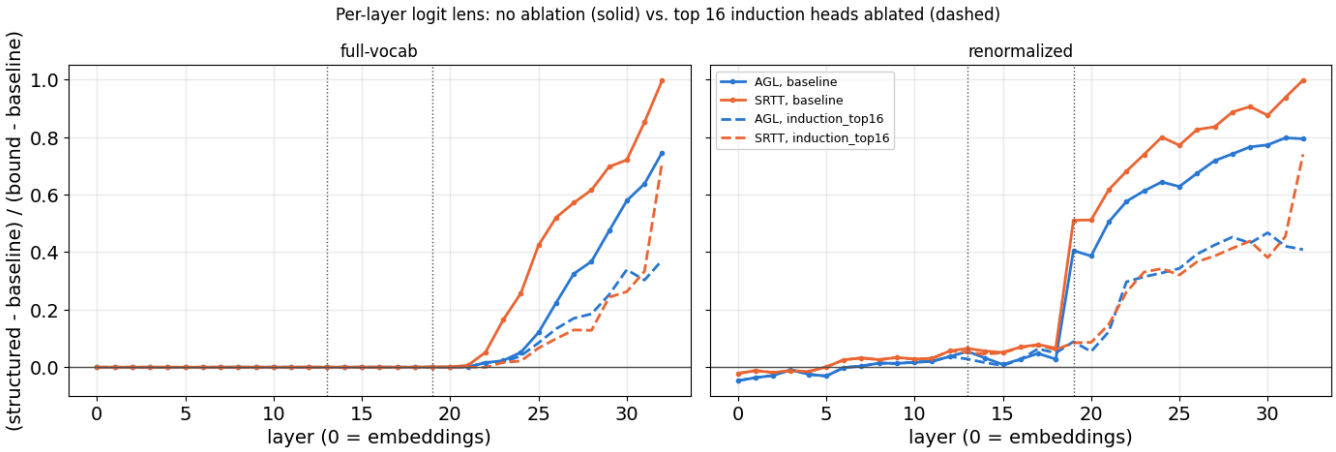
Learning curves. Without the top 16 (orange), AGL climbs to ~0.28 instead of ~0.40 and SRTT levels off at ~0.77. Random heads (blue) lie on the no-ablation curve.



Dose-response. Learning falls steadily as more heads are removed (AGL loses 21% with just 4). Wikipedia loss rises only 0.8%, so the model is not broken in general. The ICL score is too noisy to show an effect.



Rule knowledge. At q1, forbidden words normally fall to 0.03; without the heads they only reach 0.11. The model loses much of its knowledge of *what cannot come next*, not just its ability to copy.



Lens layer	AGL, no ablation	AGL, ablated	SRTT, no ablation	SRTT, ablated
13	0.055	0.028	0.065	0.051
19	**0.405**	**0.089**	**0.511**	**0.085**
24	0.644	0.328	0.800	0.343
32	0.794	0.409	0.998	0.740

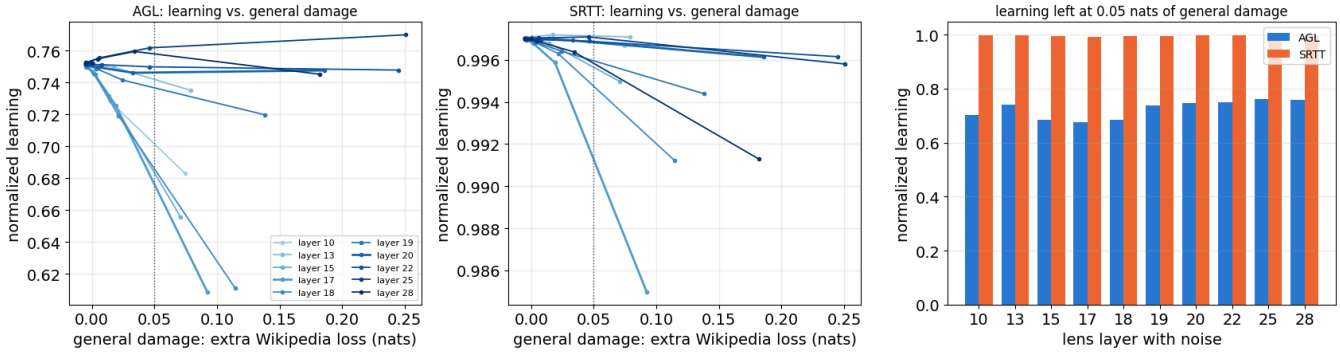
- **The jump at layer 19 is gone** for both grammars: the block-18 heads produce it.
- **Learning then builds up gradually from layer ~21** to about half the normal level: other components partly take over, slowly.
- **AGL depends on these heads more than SRTT,** against the initial guess.

Note. A first run had a bug in the copying score (normalized over all words, so every head scored ~0); its 32 mostly weak heads still cut AGL to 0.41. The corrected run gets a larger effect with half as many heads.

Test 5: Noise at each layer

Question: where in the network is the learning most fragile? **Answer:** just before block 18 (lens layers 15–18); after layer 19 noise costs almost nothing. The effect is small, but it points the same way as Test 4.

Design. At one lens layer, every word's internal state gets a random push of relative size α (0.05–0.8 of the state's own size; the start token untouched). 10 layers × 5 noise sizes, 10 paired runs per grammar, plus a general-damage check on 25 Wikipedia texts. Layers are compared **at equal general damage** (extra Wikipedia loss), so "fragile" means fragile for the learning specifically.



Lens layer with noise	10	13	15	17	18	**19**	20–28
AGL learning left at 0.05 nats of damage (no noise: 0.75)	0.70	0.74	0.68	0.68	0.69	**0.74**	0.75–0.76

- **Most fragile just before the jump:** noise on the state the block-18 heads read costs ~10% of AGL learning; right after them, nothing.
- **Late layers dissociate learning from general ability:** noise at 22–25 hurts Wikipedia most (+0.25 nats at α = 0.8), yet AGL learning does not drop.
- **SRTT is almost untouchable** (worst 0.985 vs. 0.997): too redundant for noise to break.

Caveats. Even α = 0.8 removes only ~19% of AGL learning, since random noise mostly hits directions the model does not use. Error bars show the spread of levels, not of paired differences; the damage-matched values are interpolated. The weakest of the lines of evidence.

Test 6: What each head does, per transition

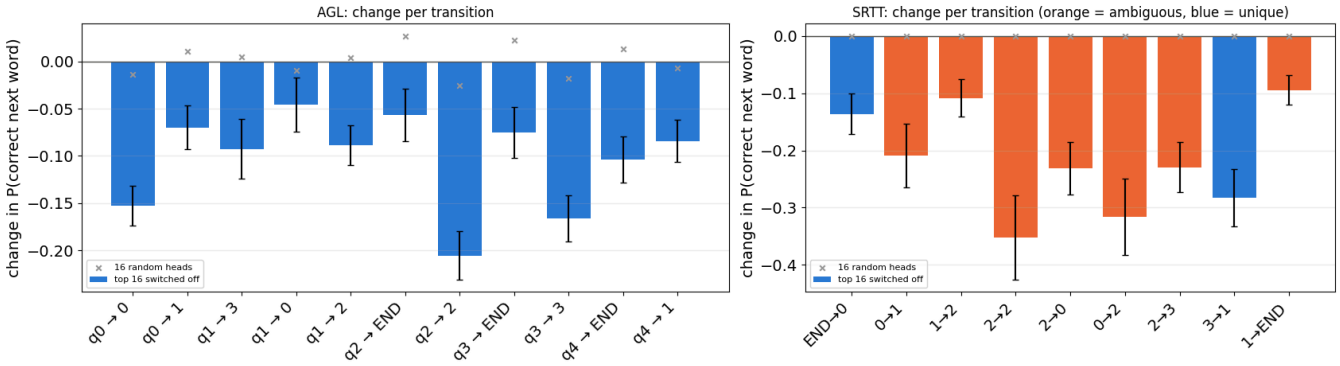
Question: does each induction head handle different transitions of the grammar?
Answer: in SRTT the block-18 heads are interchangeable copies of one mechanism, and any one nearly suffices. In AGL they are partly divided, some pushing toward END and others toward continuing, and the learning needs their combination.

Design. Learning is split by transition. SRTT: 9 positions in the cycle, 7 ambiguous (the previous word alone does not tell the next one) and 2 unique (3→1, END→0). AGL: 11 state → word transitions. Four steps on the top-16 heads (20 paired runs per grammar):

1. **Per-transition breakdown** of the top-16 switch-off from Test 4.
2. **Step A, attention:** where each head looks at each transition.
3. **Step B, single-head ablation:** switch off one head at a time.

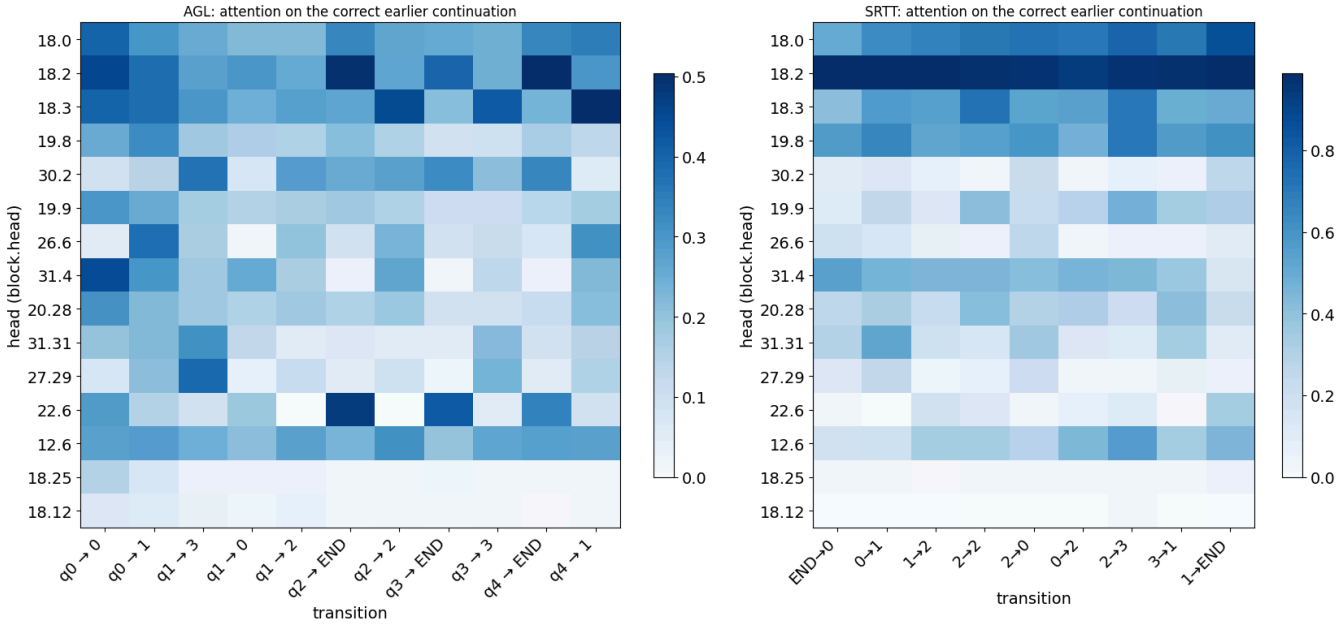
4. **Step C, leave-one-in:** switch off all 16 except one, and measure what that head restores.

Which transitions suffer when all 16 are off



- **All transitions get worse** in both grammars; random heads change almost nothing.
- **SRTT: no ambiguity effect.** Ambiguous transitions drop 0.220, unique ones 0.210 (difference −0.011 ± 0.035).
- **AGL:** hardest hit are q2→2 (−0.21), q3→3 (−0.17), q0→0 (−0.15); END transitions least (−0.06 to −0.10).

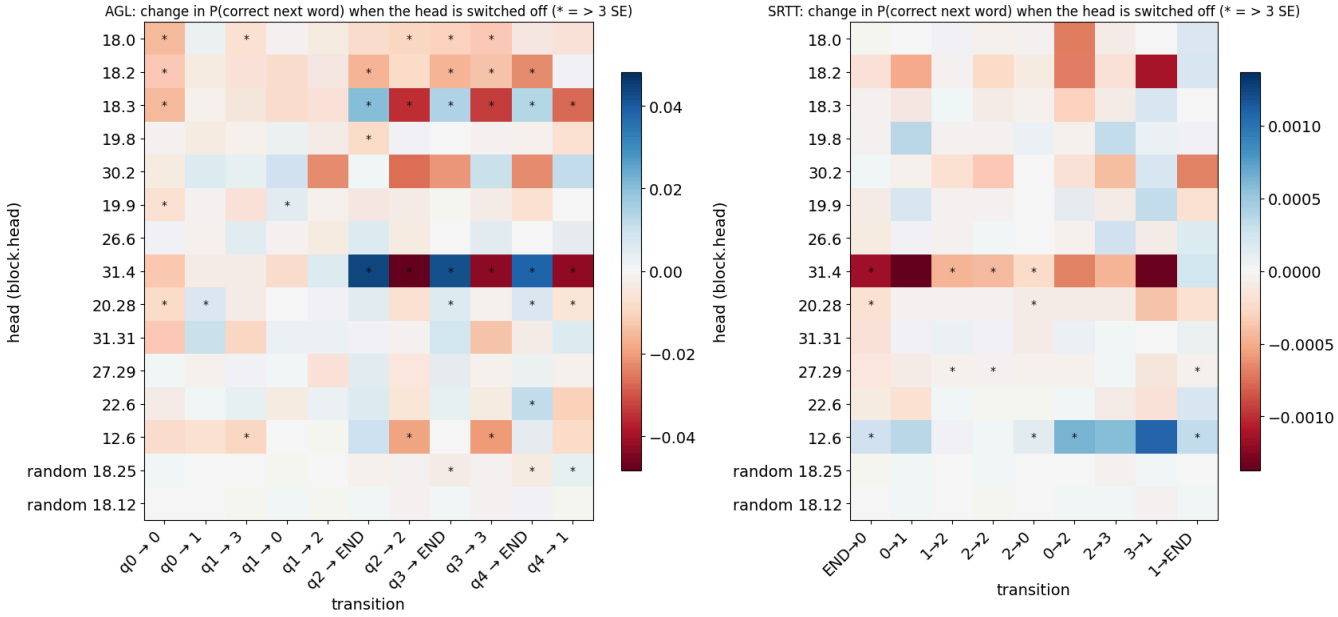
Step A: where the heads look



- **In SRTT, head 18.2 resolves the ambiguity itself:** ~97–99% of its attention goes to the correct earlier continuation at all 9 transitions, including the 7 ambiguous ones. A plain one-word induction head cannot do that, so it tracks *where in the cycle* it is.

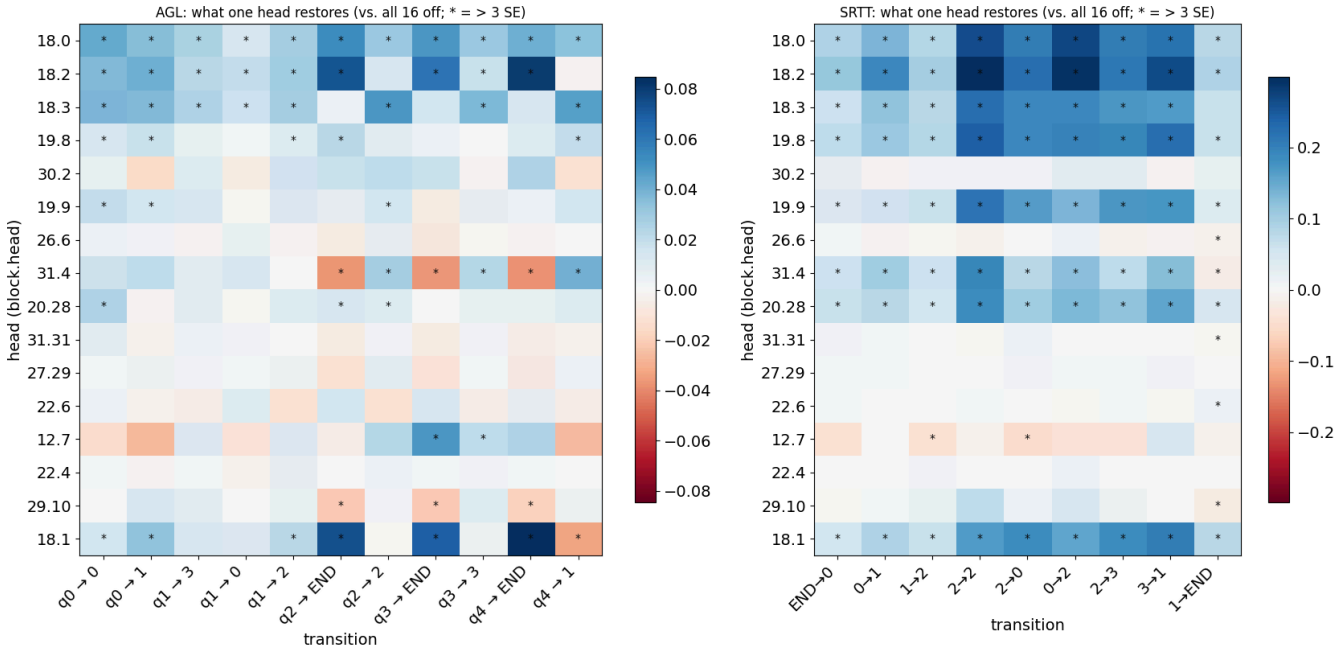
- **In AGL, the heads spread attention over all allowed continuations** (~0.25–0.3 for each of 3 options at q1, ~0.3–0.5 for 2 options elsewhere): a way to produce calibrated probabilities, not just copies.
- **Hint of a division of labour:** 18.2 attends more when the answer is END (0.40–0.50 vs. 0.25–0.30), 18.3 more when it is a word (0.42–0.50 vs. 0.22–0.26).

Step B: switching off one head at a time



- **SRTT: zero effect.** No single head moves any transition by more than 0.001.
- **AGL: small effects,** except that switching off 18.3 or 31.4 makes word continuations at q2/q3/q4 worse and END better (±0.03–0.05).

Step C: one head back on



	SRTT	AGL
Loss with all 16 off (mean)	0.216	0.104
Best single head	18.2 restores 0.198 (~92%)	18.2 restores 0.036 (~35%)
Other strong heads	18.0 ~80%, 19.8 ~71%, 18.3 ~65%, 18.1 ~61%	18.0, 18.1, 18.3: 0.027–0.035
Pattern	Even across all 9 transitions	18.1 and 18.2 restore END (+0.06 to +0.085); 18.3 restores word continuations; 31.4 the opposite of 18.1/18.2

- **SRTT is highly redundant:** one block-18 head brings back most of the learning. This is why removing the top 4 barely hurt SRTT (0.988) and why single-head ablation (Step B) had no effect.
- **AGL needs several heads together,** matching the steady dose-response of Test 4.
- **The END vs. continue split now shows in all three steps** (attention, single ablation, leave-one-in).
- **12.7 (block 12) restores almost nothing alone,** consistent with feeding block 18 rather than acting alone.

Caveats. The AGL split is about which kind of word to output (END vs. continue), not about grammar states: no head restores one state's transitions while leaving another's alone. Leave-one-in measures one head inside a damaged network. The specialization index is meaningless for heads that restore ≈0 (e.g. 12.7 scores 0.998). Stars (> 3 SE) are uncorrected over ~200 cells, so isolated ones may be chance.

Conclusions

1. **The paper's result reproduces.** A frozen Mistral-7B learns both grammars from context alone, learns which continuations are forbidden, transfers the structure to new words, and penalizes violations.
2. **The learning becomes readable abruptly at layer 19** of 32, for both grammars, with a small step at layer 13. A tuned lens shows the same, so it is a property of the model, not of how we read it.
3. **Induction heads in block 18 produce the jump.** Switching off 16 of 1,024 heads (1.6%) erases it, halves AGL learning and cuts SRTT by 28%; 16 random heads from the same blocks do nothing, and general language ability barely changes. Noise just before block 18 hurts the learning more than noise after it.
4. **The heads carry rule knowledge, not only copying:** without them the model no longer learns which AGL words are forbidden, and their attention spreads over all allowed continuations.
5. **They are not the whole story.** About half the learning remains and builds up gradually in later layers.
6. **The two grammars use the heads differently.** In SRTT any single block-18 head restores 60–90% of the loss: redundant copies. In AGL a single head restores at most ~35%; the heads combine and split between pushing toward END and toward continuing. Against the initial guess, AGL depends on the heads more than SRTT.

Limitations and next steps

- **Readable is not computed.** Both lenses are linear and aimed at next-word prediction. Next: a linear probe per layer that predicts the grammar state (q0–q4) directly, trained and tested on different runs.
- **Which cluster does what?** Switch off the block-12 cluster alone and the block-18 cluster alone, each with a random control from the same block.

- **Do the AGL heads add up?** Turn back on pairs (18.2 + 18.3) and all four block-18 heads, and check whether the END and continue restores combine.
- **How do blocks 12 and 18 interact?** Path patching (switching off only the connection between them).
- **What carries the remaining half?** Ablate 32–64 heads by the corrected ranking.
- **Other models:** repeat on Llama-3.1-8B to see whether the jump sits at the same relative depth. Untrained-model control not yet run.
- **Statistics:** p-values on the vocabulary-switch and valid/swapped plots treat neighboring positions as independent (too small); the fix is one number per run and a paired test. SRTT runs share one sequence, so their error bars are narrower. Noise-test error bars are not paired. Significance stars in Test 6 are uncorrected.
- **8-bit model** (memory limits), versus 16-bit in the paper.
- **No counting baseline** yet: a word-pair or word-triple counter would show how much of the AGL curve simple statistics explain.